ASTROPHYSICS LABORATORY
Observational Astrophysics

I) Basic notions in Observational astrophysics

a) Properties of the astronomical objects

- Magnitude; Absolute magnitude
- Magnitude and distance: ly, parsec
- Temperature and colour
- Size

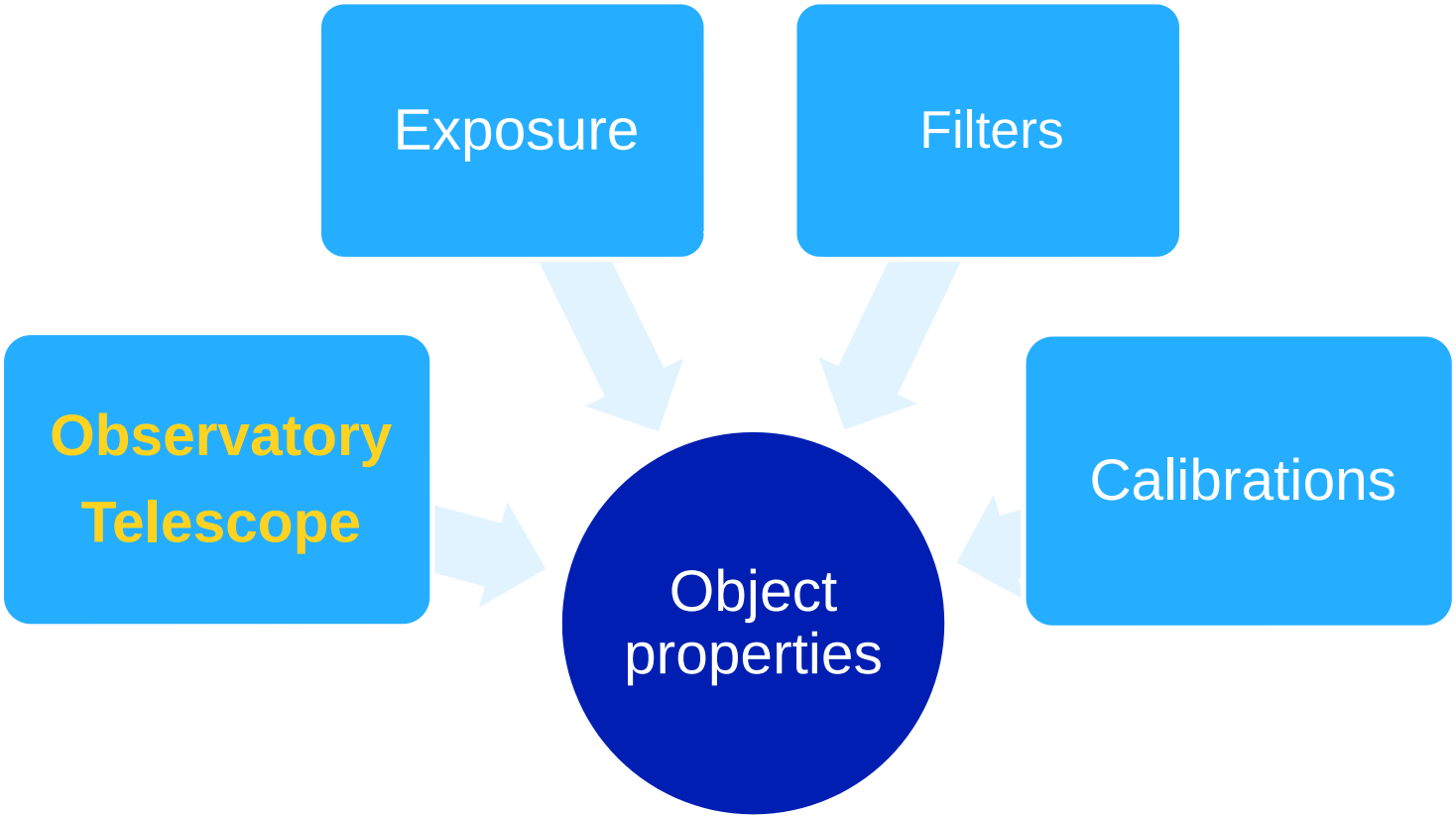
b) Preparing astronomical observations

- Visibility of the target: coordinates and time
- **Telescope**
- Detector and data acquistion

II) Observations: data acquisition

III) Data reduction and analysis

PREPARING FOR AO



OBSERVATIONS

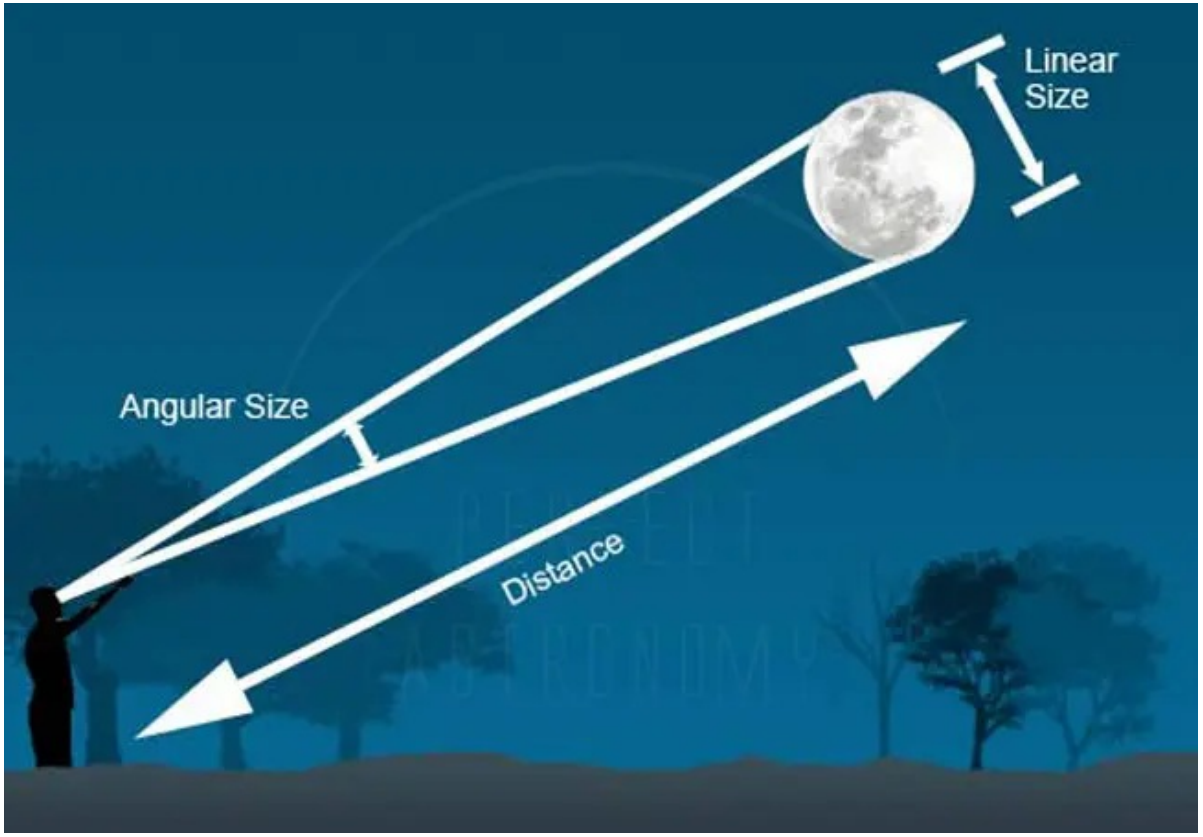
TELESCOPES

Why do we need telescopes?

1. Collect more light (photons) than human eye

2. Magnify the size of the object and see details

Angular size:
Full Moon
30arcmin = 0.5deg



Characteristics of the telescope

CELESTRON 14”

aperture: 14”= 355.6 mm

Light gathering power:

Capacity to collect photons



Curiosity

Light gathering power relative to the human pupil

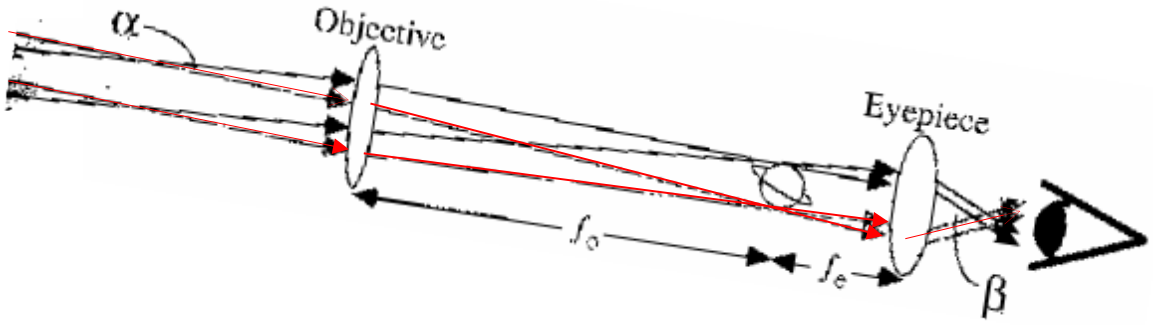
Diameter of an adult pupil: 7mm

Diameter of the C14”= 355.6 m

$$R = D^2_{telescope} / D^2_{pupil} = 2580$$

Characteristics of the telescope

magnification



magnification of the telescope: $M = f_0 / f_e = \beta / \alpha$

focal ratio: objective focal length/aperture: $N = f_0 / D$

@C14: f/11

K-Holiday, Introductory Astronomy, p 21, fig 2.6

https://www.astronomics.com/info-library/astronomical-terms/focal-ratio/

https://en.wikipedia.org/wiki/Optical_telescope#Observing_through_a_telescope

Characteristics of the telescope

magnification

CELESTRON 14”

aperture: 14”= 355.6 mm
Focal ration: N=11

f_o=N x D= 11 x 35.56=391 cm
~4000 mm

Eye pieces:
4000/40 = 100
4000/25 =160

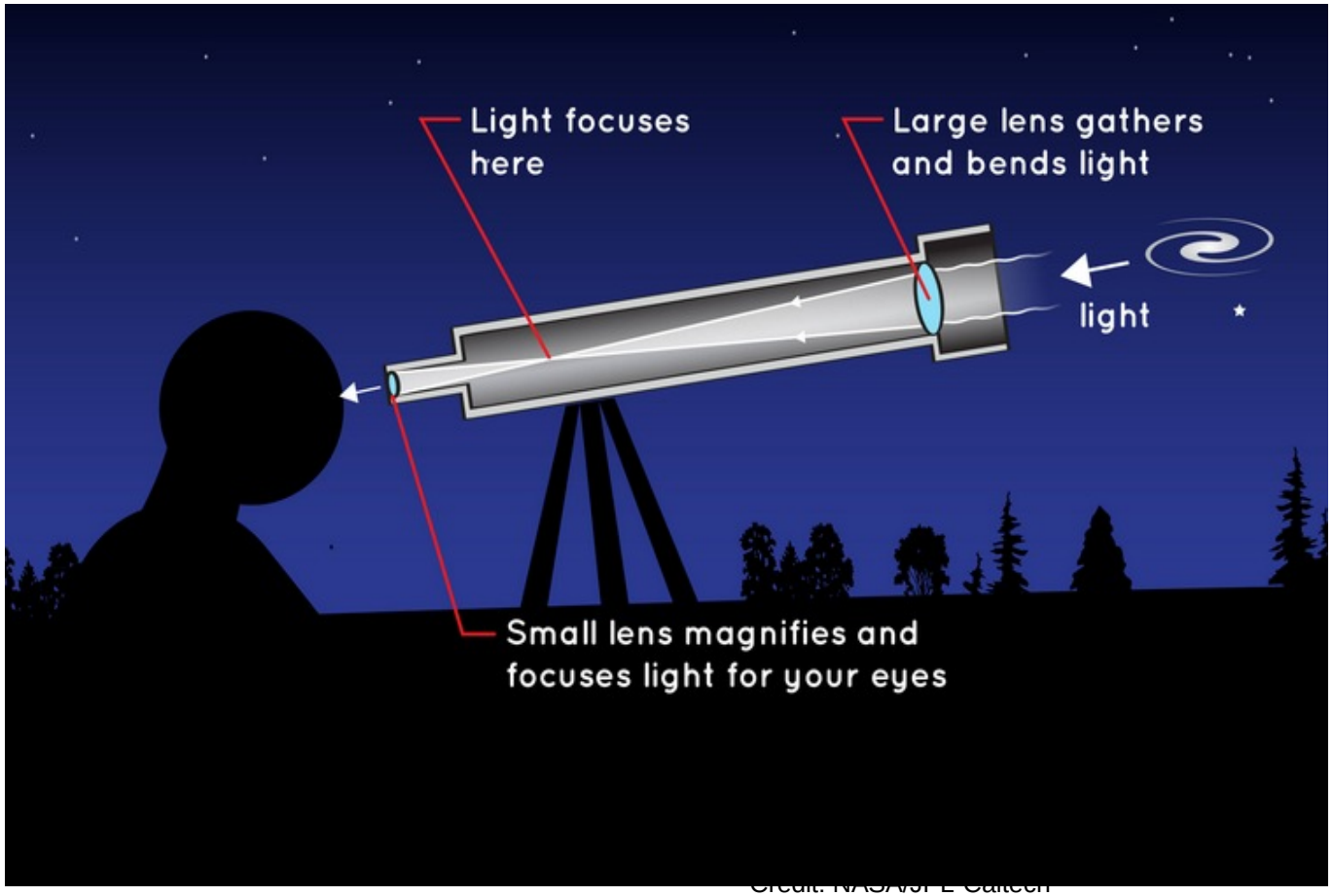


Eyepiece

lens attached to the telescope located near the focal point.

Refractor Telescope

use lenses to focus the light

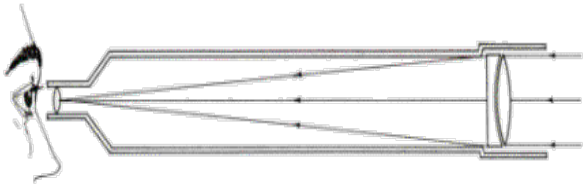


Credit: NASA/JPL-Caltech

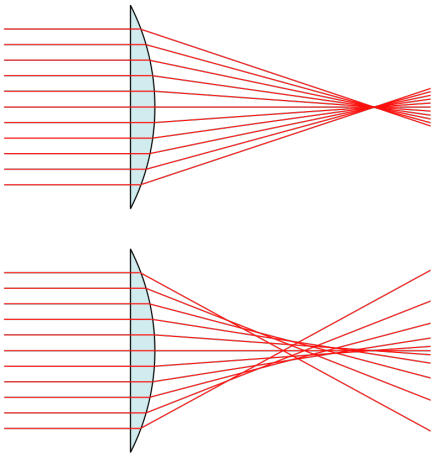
https://spaceplace.nasa.gov/telescopes/en/

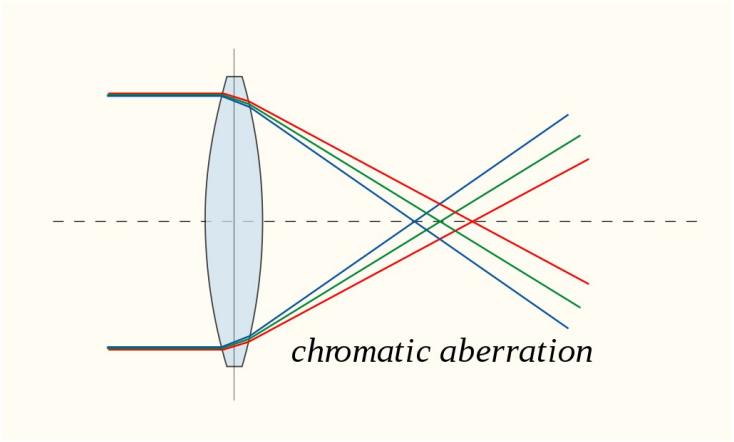
Refractor Telescope

use lenses to focus the light



Problems:
Suffer from chromatic and spherical aberration
In large apertures, lens sagging: deformation of lens by gravity

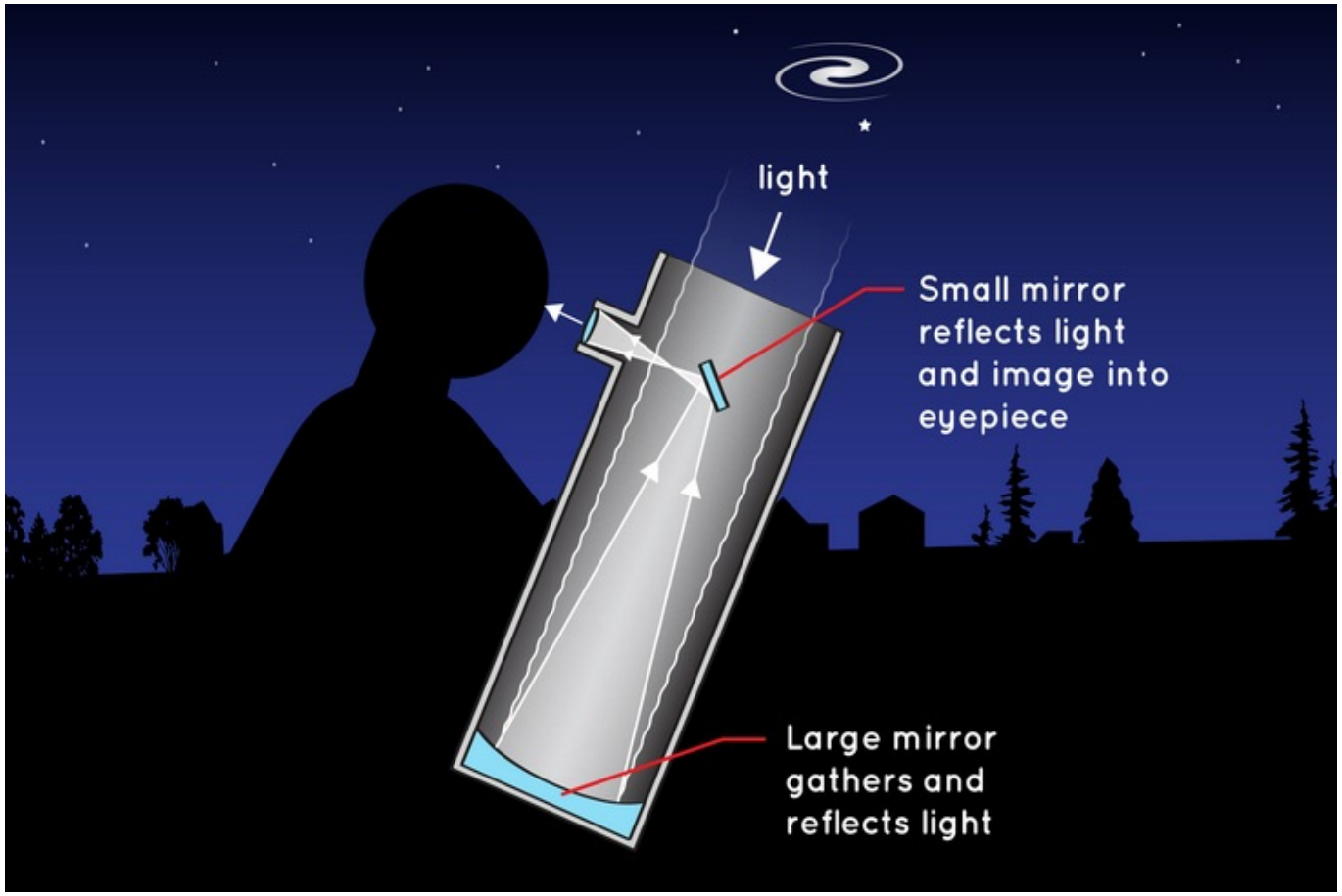




K-Holiday, Introductory Astronomy, p 21, fig 2.6
http://www.tayabeixo.org/que_obs/refractor.htm

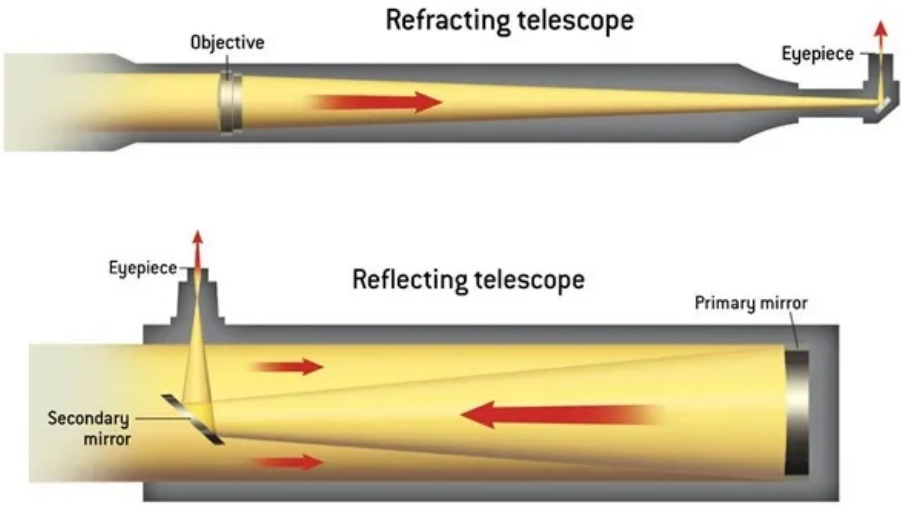
Reflecting Telescope

use mirrors to focus the light



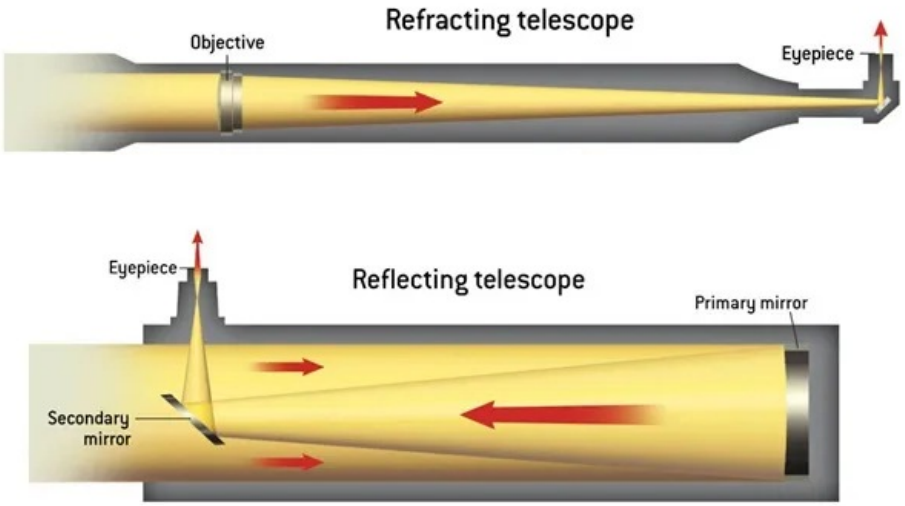
Credit: NASA/JPL-

Characteristics of the telescope



https://www.astronomy.com/observing/telescopes-101/

Characteristics of the telescope





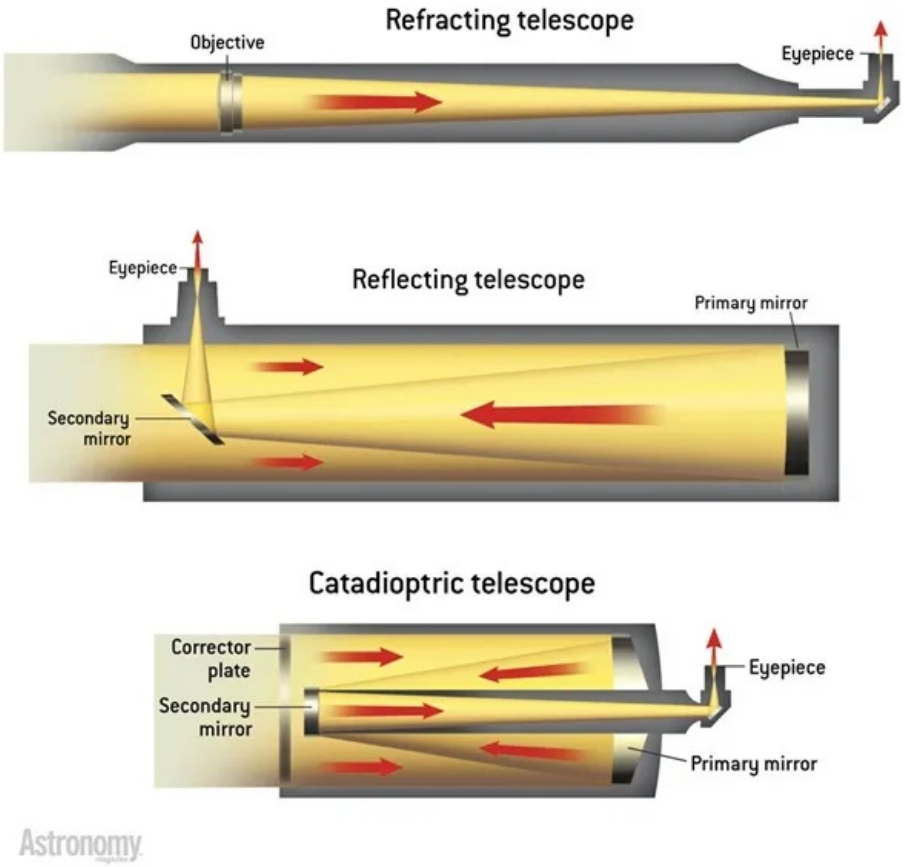


Newtonian reflector

Astronomy

https://www.astronomy.com/observing/telescopes-101/

Characteristics of the telescope



Reflectors:
use mirrors

Newtonian reflector
(parabolic+flat)

Ritchey-Chrétien
(hyperbolic+hyperbolic)

Schmidt-Cassegrain
(parabolic+hyperbolic)

https://www.astronomy.com/observing/telescopes-101/

Characteristics of the telescope





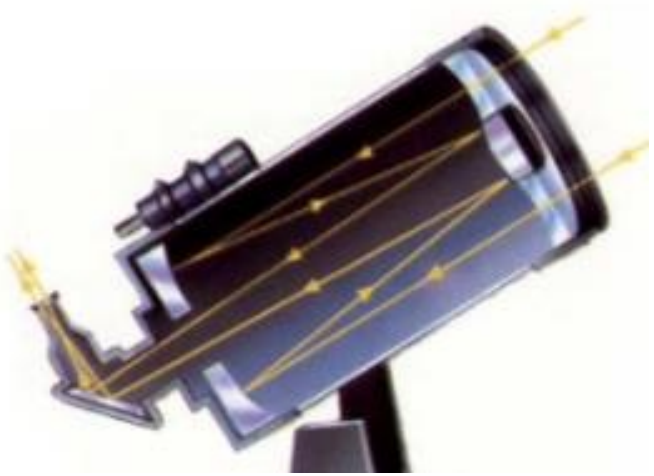


Ritchey-Chrétien

Schmidt-Cassegrain

Characteristics of the telescope

Model: Schmidt-Cassegrain



Celestron 14” catadioptric telescope
(lenses and mirrors)

Characteristics of the telescope

light-gathering power: collect light
angular resolution: see small details

magnification makes object *appear larger (angular size)*
focal length and **focal ratio**

More telescope concepts

- **field of view:** region of the sky that is observed
- **slewing**: move to a another object
- **pointing:** Point to a particular object/region
- **tracking:** Track objects on the sky
- **instrument**: detector of the light

Telescope mounts

1. **Alt-azimuth mount**: one axis for azimuth and another for altitude.
2. Simple design but complex two-axis movement to follow Earth rotation.
3. For large telescopes support goes through center of mass and there are less torques

4. **Equatorial mount (RA,DEC)**: one axis aligned to the rotational axis of Earth (celestial pole).
5. It can easily follow objects as they move through the sky

6. C14 at the CIGeoE has an Equatorial mount

Characteristics of the telescope

Angular resolution and Rayleigh criterion



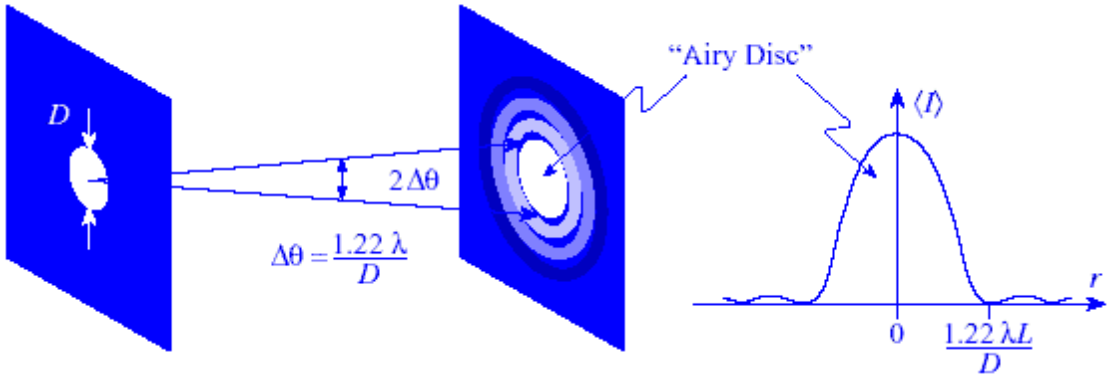
http://electron9.phys.utk.edu/optics421/modules/m5/Diffraction.htm





Characteristics of the telescope

Angular resolution

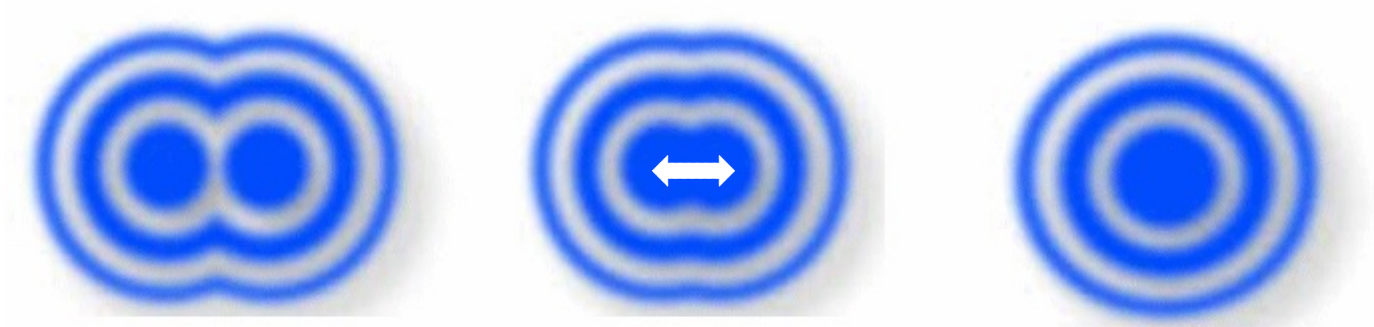


Airy disc: intensity pattern

http://electron9.phys.utk.edu/optics421/modules/m5/Diffraction.htm

Characteristics of the telescope

Angular resolution and Rayleigh criterion



well resolved just resolved not resolved

$$\theta_{min} = 1.22\,\lambda / D$$

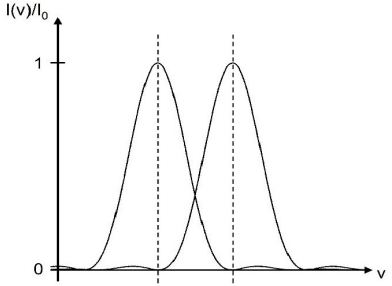
Diameter of the apperture

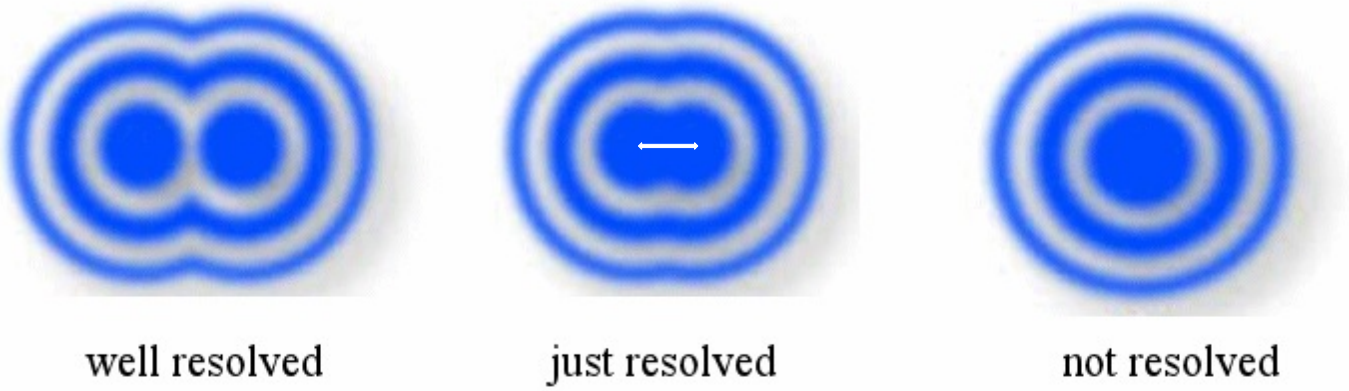
Wavelength of the observation

http://electron9.phys.utk.edu/optics421/modules/m5/Diffraction.htm

Characteristics of the telescope

Angular resolution and Rayleigh criterion





well resolved just resolved not resolved

Two point sources are regarded as just resolved when:
the principal diffraction maximum of one image coincides with the first minimum of the other

$$\theta_{min} = 1.22\,\lambda / D \;\wedge\; \theta_{min} = y / f \qquad\qquad y = 1.22\,\lambda f / D$$

http://electron9.phys.utk.edu/optics421/modules/m5/Diffraction.htm

Characteristics of the telescope

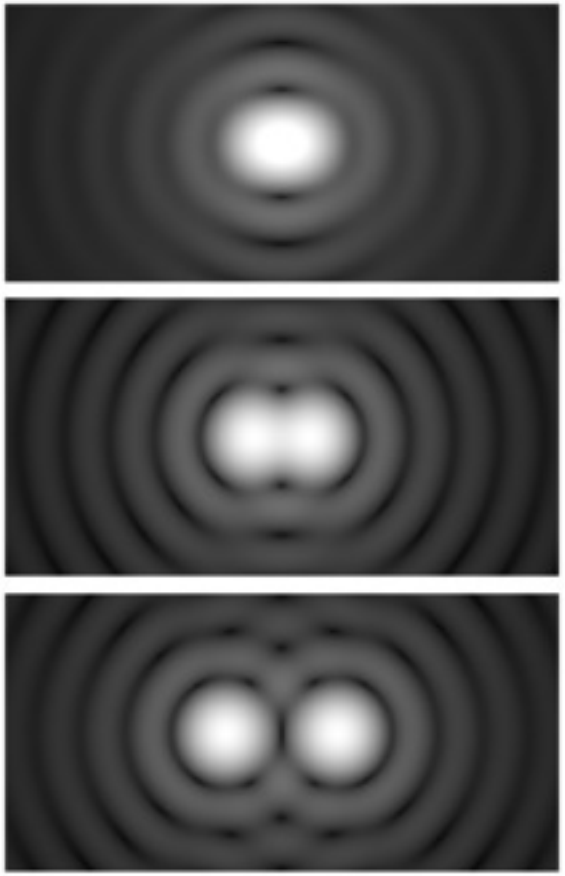
Angular resolution and Rayleigh criterion

Image resolution

ability of a telescope to distinguish

small details of an object

different objects (ex stars)



Rayleigh criterion and diffraction limits

Human eye: $D = 5\,mm, \theta_{min} \approx 27''$

Small telescope: $D = 10\,cm, \theta_{min} \approx 1.4''$

Medium telescope: $D = 1{,}0\,m, \theta_{min} \approx 0.14''$

Large telescope: $D = 8{,}0\,m, \theta_{min} \approx 0.02''$

Credi for the slide:
S. Gonzalez-Gaitán, LA@IST 2023

Characteristics of the telescope

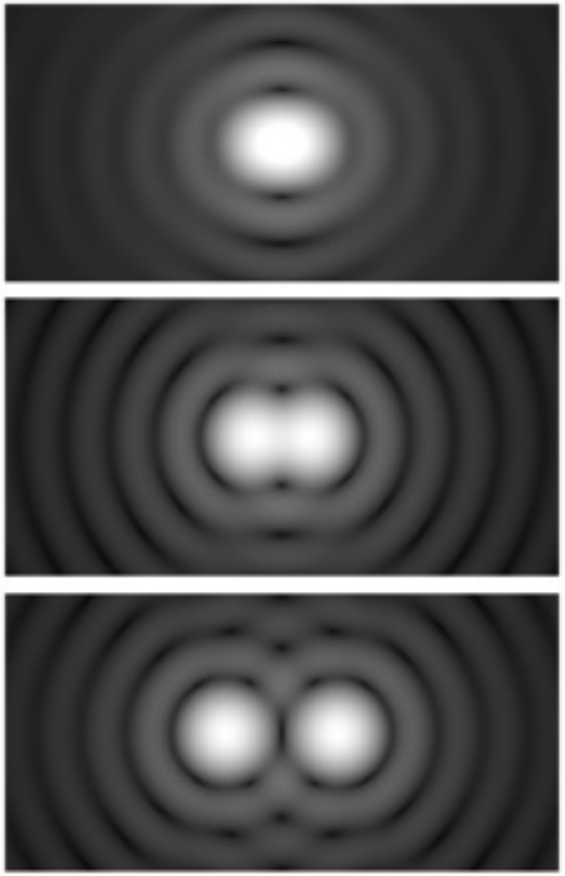
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Only true outside of the Earth's atmosphere!

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Characteristics of the telescope

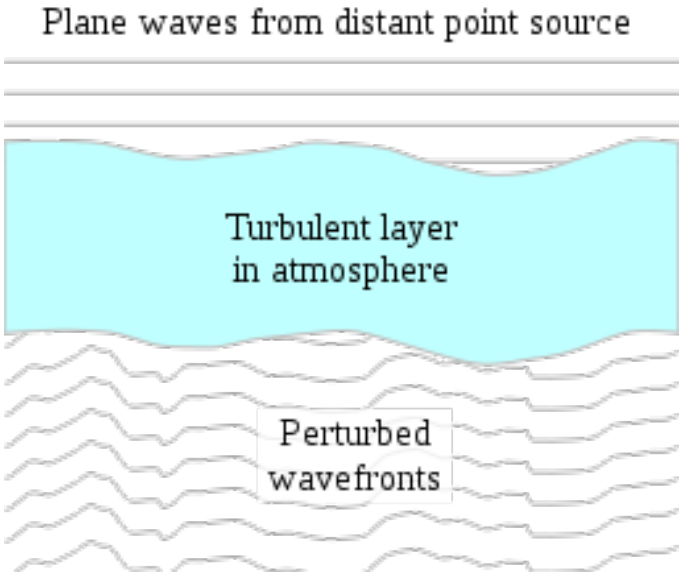


Characteristics of the telescope

In space: Rayleigh criterion applies

$$\theta_{min} = 1.22\,\lambda / D$$

On the ground: the resolution is limited by
the atmospheric turbulence (seeing),
optics, etc.





Seeing: blurring and twinkling

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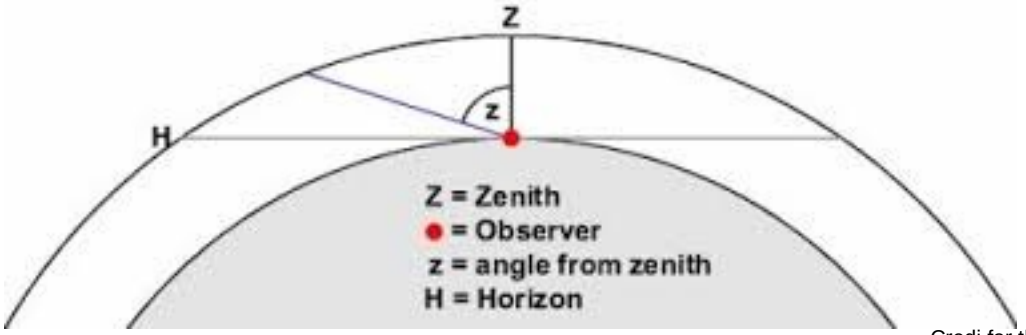
Characteristics of the observations

Seeing

Seeing is measured in arcseconds

- Seeing is **wavelength** dependent
  $\alpha_{seeing}(550nm) \approx 1''$ whereas $\alpha_{seeing}(2.2\mu m) \approx 0.75''$

- Seeing is **airmass** dependent: higher airmass means a thicker layer of atmosphere and more turbulence



Credi for the slide:
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Characteristics of the observations

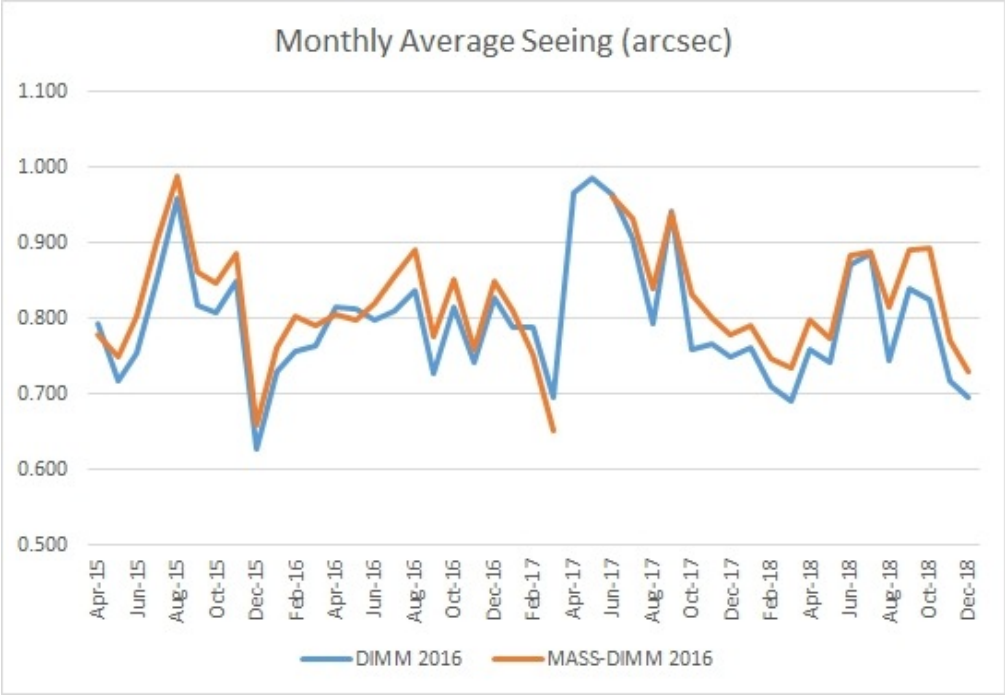
Seeing

Seeing is measured in arcseconds.

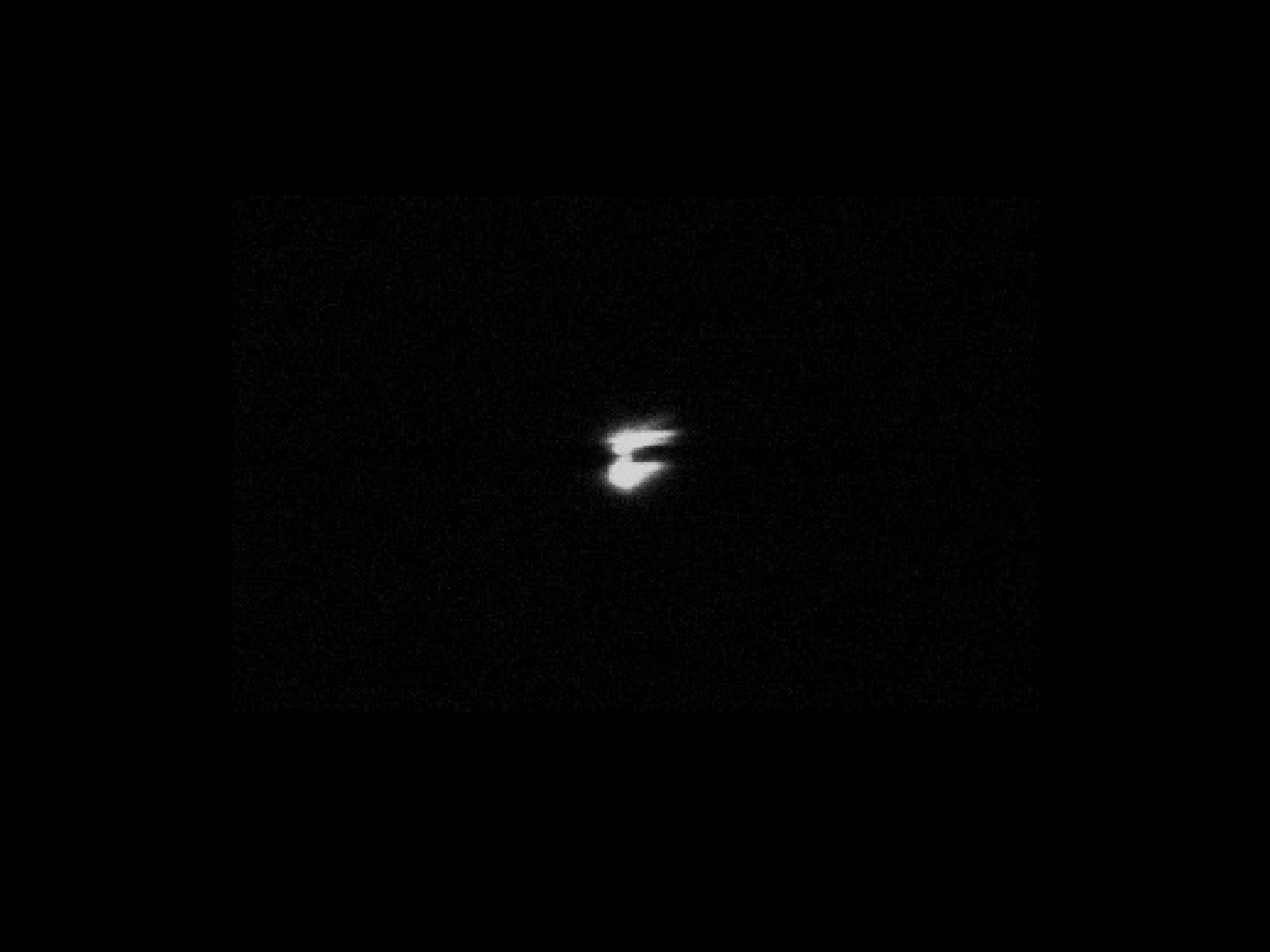
Median seeing at one of the best astronomical sites in the world is ~ 0.7’’

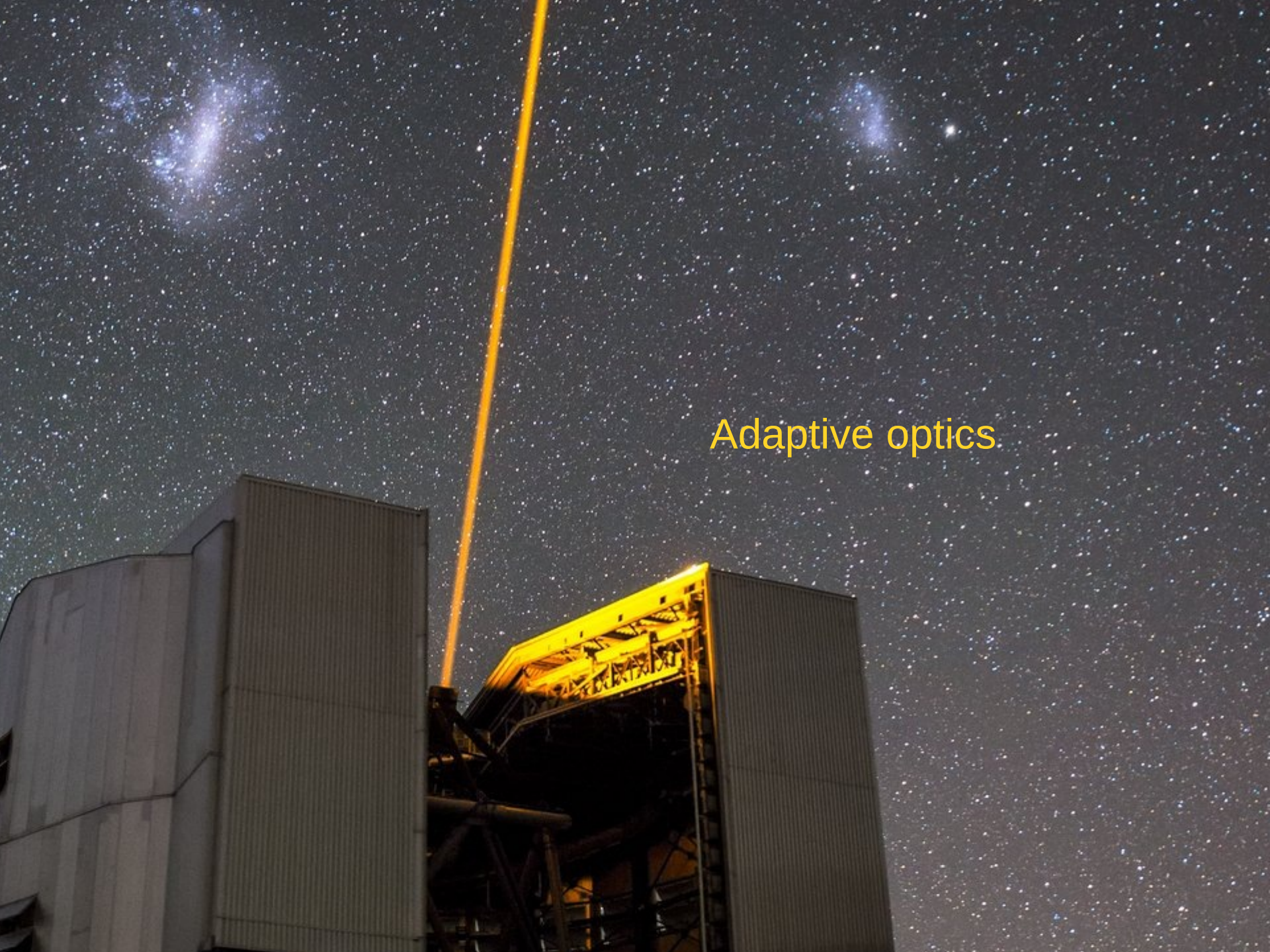
in Lisbon, seeing is >3-4’’

This is comparable to the angular resolution of a 20 cm telescope!



Credi for the slide:
S. Gonzalez-Gaitán, LA@IST 2023




Adaptive optics

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III) Data reduction and analysis

More information:

https://fenix.tecnico.ulisboa.pt/disciplinas/LA112/2025-2026/1-semestre
https://fenix.tecnico.ulisboa.pt/disciplinas/LA112/2025-2026/1-semestre/useful-tools